

Project Module 4: Privacy Risk & Privacy-Aware Design **

Objective:

In this module, you will evaluate privacy risks in your ML system, apply a simple privacy-preserving modification, and analyze the privacy–utility trade-off.

You will build on:

- Your **Module 1 dataset**
 - Your **Module 2 model**
 - Your **Module 3 analysis**
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Data and Model Reuse Requirement

For this module, you must:

- Continue using your **same dataset (German Credit or LSAC)**
 - Use the **same preprocessing pipeline**
 - Use the **same model (or similar model family)**
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What you will do

You will:

1. analyze privacy risks in your dataset and model,
 2. implement a simple privacy-preserving modification,
 3. evaluate its impact on performance and fairness,
 4. analyze privacy–utility trade-offs,
 5. propose a privacy-aware deployment design.
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** You are encouraged to use Large Language Models (LLMs) to support your thinking and learning. If you do so, you must include a **public link to the conversation(s)** used.
AUC's academic integrity rules apply: submitted work must reflect your own understanding.

Deliverables:

1. Completed notebook
 2. Short technical report
 3. Written answers must reference specific notebook outputs
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You are expected to reuse your existing code from Modules 1 and 2 and may reference, adapt, or integrate relevant code from Labs 3 and 4 and any provided assignment notebooks where appropriate.

Tasks:

Part 1: Privacy threat assessment (25%)

Identify potential privacy risks.

Answer:

1. What are the sensitive attributes in your dataset?
 2. What information could an attacker infer?
 3. Could individuals be identified or profiled?
 4. What harms could result from privacy leakage?
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Part 2: Privacy-preserving modification (30%)

Implement **one privacy intervention**, such as:

- Removing sensitive features
- Coarsening features (e.g., age → age groups)
- Reducing number of features (data minimization)
- Removing prediction probabilities (output minimization)

Evaluate the model after modification.

Include:

- Implementation code
- Performance results

Part 3: Utility and fairness impact (20%)

Compare:

- Accuracy before vs after
- Fairness metrics
- Feature importance changes

Include:

- Comparison table
- At least one visualization

Answer:

1. Did performance decrease?
 2. Did fairness improve or worsen?
 3. Which features became more/less important?
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Part 4: Privacy vs utility analysis (15%)

Answer:

1. What trade-offs did you observe?
 2. Why does improving privacy reduce model utility?
 3. Is the trade-off acceptable in your application?
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Part 5: Privacy-aware deployment design (10%)

Design a simple deployment plan.

Include:

- What data should be collected?
- What should not be collected?
- Who can access outputs?
- Data storage and retention
- Response to privacy breaches